For a well-designed SAR ADC, quantization noise should typically dominate over other noise sources, of which thermal noise (kT/C) is the most significant. To maximize the performance of the ADC and preserve the Effective Number of Bits (ENOB), we must ensure that $kT/C \ll P_q$ (Quantization Noise Power).

Note: For these derivations and calculations, a single-ended ADC model is used. The differential version can be directly inferred from these derivations.

1. Thermal Noise Model

The noise power introduced across the DAC capacitor by the sampling resistor can be modeled as:

$$P_{tn} = 4kTR \cdot B$$

Where R is the total noise-generating resistance, and B is the noise equivalent bandwidth. Due to the RC low-pass filter nature of the sampling network, the noise equivalent bandwidth is:

$$B = \frac{1}{2\pi RC}$$

Thus, the total integrated thermal noise power becomes:

$$P_{tn} = \frac{4kTR}{2\pi RC} = \frac{2kT}{\pi C}$$

If we set an upper limit to kT/C, then the total sampling thermal noise will be bounded. Hereafter, we will refer to this thermal noise power as $P_{kt/c} = kT/C$.

2. SNR and ENOB Degradation

For an ideal quantizer, the Signal-to-Noise Ratio (SNR) in dB is governed by:

$$\mathrm{SNR}_{ideal}(\mathrm{dB}) \approx 6.02 \cdot N + 1.76$$

(where N is the number of bits)

When other noise sources are considered, the overall SNR decreases, resulting in a lower ENOB:

$$\mathrm{SNR}(\mathrm{dB}) = 10 \log_{10} \left(\frac{P_{sig}}{P_{tot}} \right)$$

For an ideal quantizer, $P_{tot} = P_q$. Now, assume additional noise sources are introduced and can be represented as a scaled factor a of P_q:

$$P_{other} = P_{kt/c} + P_{comp} + P_{switch} = a \cdot P_q$$
$$P_{tot} = P_q + a \cdot P_q = (1 + a) \cdot P_q$$

The degraded SNR and ENOB reduction are then:

$$\mathrm{SNR}_{degraded}(\mathrm{dB}) = \mathrm{SNR}_{ideal}(\mathrm{dB}) - 10\log_{10}(1 + a)$$
$$\mathrm{ENOB}_{reduction} = \frac{10\log_{10}(1 + a)}{6.02} \approx 1.66\log_{10}(1 + a)$$

a (Noise Ratio)	ENOB Reduction (Bits)
0.01	0.007
0.10	0.070
0.20	0.130
0.40	0.240
0.50	0.290
1.00	0.500
2.00	0.790
3.00	1.000

3. Quantization Noise Power (P_q)

$$P_q = \frac{\mathrm{LSB}^2}{12} = \frac{\left(\frac{V_{ref}}{2^N}\right)^2}{12} = \frac{V_{ref}^2}{12 \cdot 2^{2N}}$$

For our specific design ($V_{ref} = 1.0\ \mathrm{V}$, $N = 8$ bits):

$$P_q = \frac{1^2}{12 \cdot 2^{16}} = \frac{1}{786432} \approx 1.2716 \times 10^{-6}\ \mathrm{V}^2$$

4. Capacitor Sizing for Thermal Noise

Assuming the dominant extra noise source is kT/C (so $a \approx A$), considering the maximum operating temperature of 125°C (398.15 K) ensures the inequality holds for all lower temperatures:

$$\frac{kT}{C} < a \cdot 1.2716 \times 10^{-6}\ \mathrm{V}^2$$

$$\frac{5.494 \times 10^{-21}}{C} < a \cdot 1.2716 \times 10^{-6}$$

$$C > \frac{4.32 \times 10^{-15}}{a}\ (\mathrm{F})$$

a (Noise Ratio)	Min Total C_DAC	Min Unit Cap (C_DAC / 2^8)
0.01	432.0 fF	1.688 fF
0.10	43.2 fF	0.169 fF
1.00	4.3 fF	0.017 fF

Conclusion: *Quantitative analysis shows that the minimum capacitance required to meet thermal noise constraints is significantly smaller than practical, manufacturable capacitor sizes for an 8-bit architecture. Therefore, capacitor sizing for this ADC is not limited by kT/C noise, but rather bounded by matching limitations, settling time, and parasitic considerations.*

5. Capacitor Array Mismatch Analysis (Pelgrom's Model)

To avoid linearity degradation (INL/DNL), matching between adjacent unit capacitors (C_u) must strictly adhere to Pelgrom's mismatch model:

$$\sigma\left(\frac{\Delta C}{C}\right) = \frac{A_c}{\sqrt{\mathrm{Area}}}$$

5.1 SPICE Model Extraction for Sky130 MiM Capacitors

We target the Metal-Insulator-Metal (MiM) capacitor devices available in the `sky130_fd_pr` library (`sky130_fd_pr__cap_mim_m3_1` / `_2`). The PDK SPICE model defines matching distributions as follows:

```
.param carea = 'camimc*(wc)*(lc)'
.param cperim = 'cpmimc*((wc)+(lc))*2'
.param czero = 'carea + cperim' dev/gauss='0.01*2.8*(carea + cperim)/sqrt(wc*lc*mf)'
```

Extracting the Gaussian deviation parameter yields the Pelgrom area coefficient: $A_c = 0.01 \cdot 2.8 = 0.028$ (or $2.8\% \cdot \mu\mathrm{m}$).

- **Area Capacitance (C_{area})**: $2.0\ \mathrm{fF}/\mu\mathrm{m}^2$
- **Perimeter Capacitance (C_{perim})**: $0.19\ \mathrm{fF}/\mu\mathrm{m}$

5.2 Sizing for 8-Bit Accuracy

To preserve half-LSB accuracy, a common worst-case design constraint requires standard deviation matching limits to strictly bound DNL errors:

$$\sigma\left(\frac{\Delta C}{C_u}\right) < \frac{1}{2^{N+1}} = \frac{1}{2^9} \approx 1.953 \times 10^{-3}$$

$$\frac{0.028}{\sqrt{\text{Area}}} < 1.953 \times 10^{-3} \implies \text{Area} > 205.5\ \mu\text{m}^2$$

Assuming a square layout (W = L) to minimize perimeter variations: $W = L \approx 14.33\ \mu\text{m}$

$$C_{unit} = (2.0 \cdot 205.5) + (0.19 \cdot 4 \cdot 14.33) \approx 422\ \text{fF}$$

5.3 Area Feasibility Analysis

A standard binary-weighted Capacitor DAC for an 8-bit differential architecture requires 512 total unit capacitors (256 per side).

$$\text{Total Active Cap Area} = 512 \cdot 205.5\ \mu\text{m}^2 \approx 105,216\ \mu\text{m}^2\ (\backslash\text{sim } 105\text{k}\ \mu\text{m}^2)$$

Note: This calculation ignores dummy capacitors, routing density, and spacing.

- Required Area (Array Only): **> 105,000 µm²**
- Available Area (Total 1x2 Tiles): **36,000 µm² (160 x 225 µm)**

6. Architecture Implications

The mismatch requirement strictly prohibits the implementation of a standard full-binary CDAC within the allocated TinyTapeout area constraints. To achieve the 8-bit core (and 10-bit overall ENOB via noise shaping) in the required footprint, an aggressive area-reduction strategy is mandatory.